Modeling Correlated-Data in Sensor Networks using PCA and Kernel Regression

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I. Introduction

A Wireless sensor network (WSN) typically comprises a large number of tiny, battery-operated nodes that are spatially-distributed and networked to cooperatively collect, process, and deliver data to a *sink* about a phenomenon of interest. When sensor measurements are correlated over space and/or time, which is often the case in such networks, data is *aggregated* using simple local computations (e.g., AVG, MIN, MAX, etc.) en route to the sink. Although aggregation conserves energy and eliminates redundancy, it can lose much of the original structure in data, providing only coarse statistics that smooth over interesting local variations.

In this project, we explore two machine learning techniques, *principal component analysis* (PCA) and *kernel regression* (KR), that can extract much more complete information about the shape and structure of data than most aggregation schemes while consuming less energy. In particular, we use PCA to eliminate spatial correlation in data from different sensors by projecting onto a lower dimensional subspace spanned by the principal components [1]. To model temporal correlation in data from a particular sensor, we use kernel regression [2] to approximate measurements by using a weighted linear combination of basis functions, called *kernel functions*.

Let p be the total number of nodes deployed in a two-dimensional terrain with (x_i, y_i) denoting the coordinates of node i. Let $\mathcal{D} = \{f(x_i, y_i, t) : i = 1, \ldots, p;\ t = 1, \ldots, N\}$, be the set of sensor readings collected over total time N. Note that, we consider the measurements $f(.)$ as a function of both location and time in order to model spatio-temporal correlations. For notational convenience, we use $f_i(t)$ in place of $f(x_i, y_i, t)$ to denote the measurement of node i at time t, and $\mathbf{f}(t) = \{f_1(t), \ldots, f_p(t)\} \in \mathbb{R}^p$ to denote the p-dimensional vector of measurements at time t. Finally, let $\bar{\mathbf{f}}(t) = \frac{1}{N}\sum_{t=1}^{N}\mathbf{f}(t)$ be the mean vector over time. Abusing notations slightly, we will also denote by $\mathbf{f}(t)$ the mean centered measurements, i.e., the mean $\bar{\mathbf{f}}(t)$ has been subtracted from the data.

Our first goal in eliminating spatial redundancy using PCA is to find a set of q orthonormal basis vectors $\{\mathbf{u}_k\}_{1\le k\le q}$ in a lower dimensional space $\mathbb{R}^q$ where $q \le p$, such that the reconstruction error is minimized. Our performance metric for PCA is the proportion of *retained variance* and the *reconstruction errors* on test data. To eliminate temporal correlation and to accurately predict unseen data, our goal is to fit a non-linear curve using kernel regression.

II. Methods

A. Principal Component Analysis

In PCA, we aim to find q basis vectors $\{\mathbf{u}_k\}_{1\le k\le q} \in \mathbb{R}^q$, such that the mean squared distance between the centered data $\mathbf{f}(t)$ and their projections $\widehat{\mathbf{f}}(t) = \sum_{k=1}^{q}\mathbf{u}_k\mathbf{u}_k{}^{\mathrm{T}}\mathbf{f}(t)$ on the subspace spanned by the basis vectors is minimized. Thus, the objective function to be minimized is

$$J_q = \frac{1}{N}\sum_{t=1}^{N}||\mathbf{f}(t) - \sum_{k=1}^{q}\mathbf{u}_k\mathbf{u}_k{}^{\mathrm{T}}\mathbf{f}(t)||^2 \tag{1}$$

under the orthonormality constraint of the basic vectors. The objective function equivalently maximizes the projected variance. Using Lagrange multipliers and setting the derivative of J_q with respect to $\mathbf{u}_k$ to zero gives

$$\mathbf{S}\mathbf{u}_k = \lambda_k\mathbf{u}_k \tag{2}$$

where $\mathbf{S} = \frac{1}{N}\sum_{t=1}^{N}\mathbf{f}(t)\mathbf{f}(t)^{\mathrm{T}}$ is the data covariance matrix. The minimizer of (1) thus comprises the q eigenvectors, called the *principal components*, corresponding to the q *largest* eigenvalues of $\mathbf{S}$. The sum of these q largest eigenvalues is the variance retained in the projected subspace. Thus, the proportion $P(q)$ of variance retained is

$$P(q) = \frac{\sum_{k=1}^{q}\lambda_k}{\sum_{k=1}^{p}\lambda_k}. \tag{3}$$

In our implementation, we use *singular value decomposition* (SVD) of the covariance matrix as $\mathbf{S} = \mathbf{U}\mathbf{D}\mathbf{V}^{\mathrm{T}}$ to find the eigenvalues and eigenvectors. Here $\mathbf{D}$ is a diagonal matrix containing the eigenvalues in decreasing order, and $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices, all of size $p \times p$. We also use a 10-fold *cross validation* to ensure that the measurements used to compute the principal components are not used for assessing the accuracy.

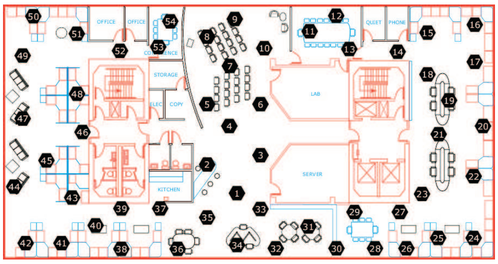


Fig. 1: Deployment at Intel-Berkeley lab with 54 Mica2Dot sensors.

B. *Kernel Regression*

In doing kernel regression over *time* for a set of measurements of a given sensor i, we fit a non-linear curve and estimate the target value $\widehat{f}_i(t')$ for a test input t' as a weighted average of the function values at the training inputs given by

$$\widehat{f}_i(t') = \frac{\sum_{t \neq t'} f_i(t) k(t, t')}{\sum_{t \neq t'} k(t, t')} \qquad (4)$$

where $k(t, t') \geq 0$ is a kernel function that decays rapidly with squared distance $d(t, t')$ between t and t'. This is to ensure that the estimated value has the strongest dependence on nearby training points. In our implementation, we use a Gaussian kernel of the following form

$$k(t, t') = e^{-\frac{d(t,t')}{2\sigma^2}} \qquad (5)$$

where σ^2 is called the *kernel bandwidth* that acts as a smoothing parameter. We can imagine the kernel bandwidth as the width of a window centered at a training input, and give weighing value to any points located in the window.

III. Experimental Results

Our data set comprises five consecutive days of temperature and voltage readings sampled every 30 seconds from 54 Mica2Dot sensors deployed in the Intel research laboratory in Berkeley between February 28th and April 5th, 2004 [3]. The deployment topology is shown in Fig. 1. The total number of readings over 5 days is $5 \times 24 \times 60 \times 2 = 14400$ from 52 different sensors (sensors 5 and 15 did not provide any measurement).

We show a sample temperature profile of sensor 1 over 5 days in Fig. 2(a). Clearly, the profile exhibits a similar pattern every day. Fig. 2(b) shows a zoomed-in plot over 24 hours (86400 seconds) starting from midnight. We observe that the temperature first drops to a minimum at around 21500 seconds (dawn), and then gradually rises reaching a maximum around late morning. Then, it remains steady between 22-24°C throughout the afternoon before slowly dropping in the evening and night. This illustrates temporal correlation in the sensor readings. Fig. 2(c) shows a scatter plot of temperatures from sensors 20 and 21 over two days. We observe that the readings are strongly correlated due to spatial proximity of the sensors (both are located to the extreme right of the floor plan in Fig 1). On the other hand, readings from sensor 21 and 49, which are located far away from each other, exhibit least correlation, as shown in Fig. 2(d).

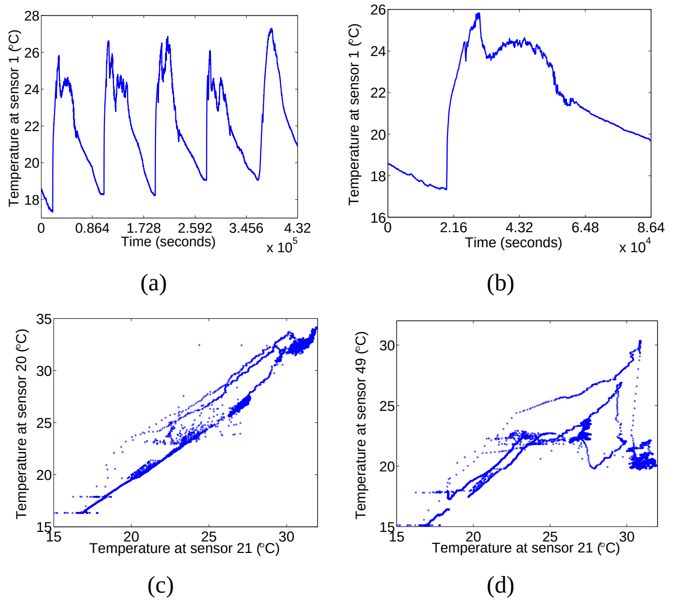


Fig. 2: (a) Temperature profile of sensor 1 over a five day period. (b) Zoomed-in plot over 24 hours for sensor 1. (c) High correlation between sensor 21 and 20 due to spatial proximity. (d) Least correlation between sensor 21 and 49.

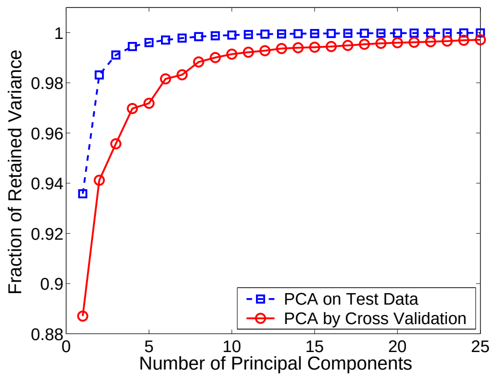


Fig. 3: Variance retained by principal components.

A. *PCA Results on Temperature*

We use a 10-fold cross validation by splitting the data into ten equal parts, each part containing 1440 readings. We compute the eigenvalues and eigenvectors by training over the nine parts, and use the remaining part to estimate the proportion of retained variance. Then we take the average of these ten estimates.

The dotted curve in Fig. 3 shows the average amount of variance retained, as computed by (3), on the test data when the principal components are also computed on the test data, whereas the solid curve corresponds to when the principal components are computed on the training data. The dotted curve therefore gives a bound on the compression efficiency of PCA on the test data. We observe that the amount of retained variance initially increases very rapidly, with the first principal component accounting for more than 88%, and then it becomes almost linear after 10 principal components. This reflects the fact that the remaining components are no better than random components, and so the remaining variations can be considered as white noise. Thus, PCA is able to reduce the dimensionality from $p = 52$ to $q = 10$ while retaining more than 98% of the variance.

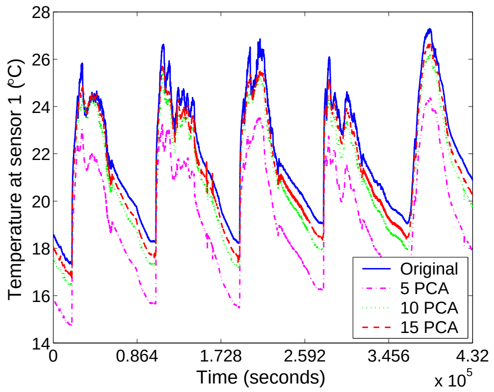


Fig. 4: Approximations obtained on sensor 1 using 5, 10, and 15 principal components.

Fig. 4 shows the approximations obtained on the test data for 5, 10, and 15 principal components. We observe that approximations improve with more principal components, however, the improvements come with diminishing returns as expected, and underlines the effectiveness of PCA is modeling spatial correlation.

B. *Kernel Regression Results*

Fig. 5 illustrates the results of kernel regression using Gaussian kernel of the form (5) for $\sigma^2 = 500$ and only 28 basis functions. The solid curve shows the actual temperature readings from sensor 49 over a period of 24 hours, whereas the dotted curve represents the non-linear fit by training over only 1% of total data. The close matching between the curves illustrates the temporal correlation in measurements from a given sensor, and the ability of KR in accurate prediction.

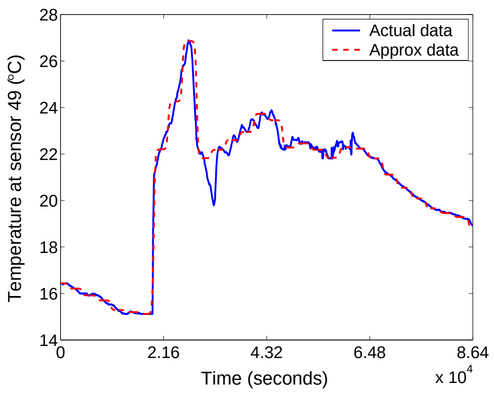
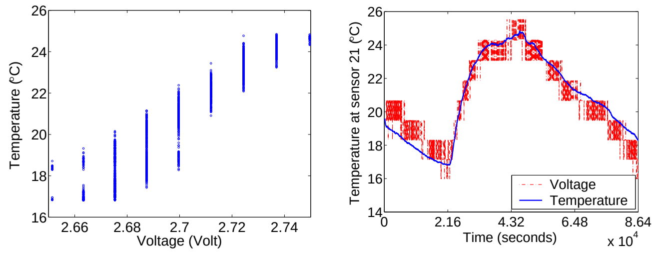


Fig. 5: Approximate temperatures by Gaussian kernel regression.



(a) (b)

Fig. 6: (a) Strong correlation between temperature and voltage. (b) Linear regression with model parameters $w_0 = -241.5, w_1 = 97.1$.

In addition to spatial and temporal correlation, we also found that there exists a strong correlation between voltage and temperature, as shown by an example in Fig. 6(a). Thus, voltages can be used to predict temperatures. Since sampling a voltage on current sensor nodes requires several orders of magnitude lesser energy than sampling a temperature [4], one can fit a linear regression, as shown by an example in Fig. 6(b) for sensor 21, and transmit only the model parameters, thus saving energy.

IV. Conclusions

We studied two machine learning techniques, PCA and kernel regression, to model correlated data in sensor networks. We found that PCA can model spatial correlation by projecting data onto a lower dimensional subspace where the measurements are uncorrelated. Thus, only the principal components in that subspace can be transmitted instead of all data for energy efficiency. We also found that kernel regression can be effectively used in modeling temporal correlation and prediction. Lastly, we explored other types of correlation, such as between voltage and temperature, and showed an example of linear regression by which once can save energy by transmitting only the voltage readings and the regression model parameters.

References

[1] Y. Le Borgne, S. Raybaud, and G. Bontempi, "Distributed principal component analysis for wireless sensor networks," *Sensors*, vol. 8, no. 8, pp. 4821–4850, 2008.
[2] C. Guestrin, P. Bodik, R. Thibaux, M. Paskin, and S. Madden, "Distributed regression: an efficient framework for modeling sensor network data," in *IPSN '04*, Berkeley, CA, pp. 1–10.
[3] "Intel lab data webpage," http://db.csail.mit.edu/labdata/labdata.html.
[4] A. Deshpande, C. Guestrin, S. R. Madden, J. M. Hellerstein, and W. Hong, "Model-driven data acquisition in sensor networks," in *VLDB '04*, pp. 588–599.